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Antimicrobial potential of TiO₂ nanoparticles against MDR *Pseudomonas aeruginosa*

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The present study aims to evaluate the inhibition effectiveness of titanium dioxide nanoparticles in combination with cell wall active antibiotics – ceftazidime and cefotaxime against the multi-drug resistant *Pseudomonas aeruginosa* isolated from pus, sputum, endo-tracheal tract and broncho-alveolar lavage. Commercial Degussa-P25 TiO₂ nanoparticle, antibiotics ceftazidime and cefotaxime were used in this study against multi-drug-resistant nosocomial pathogen. The nanoparticle shows antimicrobial effect on the pathogen at concentrations more than 350 µg/mL, when exposed to ultraviolet (UV) light for an hour. Minimum inhibitory concentration values obtained for the antibiotic cefotaxime were sixfolds higher than the antibiotic ceftazidime. When these antibiotics were used in combination with UV-irradiated metal nanoparticle, ceftazidime resulted in enhanced antimicrobial activity whereas cefotaxime does not show any change.

Keywords: *Pseudomonas aeruginosa*; TiO₂ nanoparticle; antimicrobial activity; multi-drug resistant

1. Introduction

Infections caused by multi-drug resistant (MDR) strains *Pseudomonas aeruginosa* are becoming extremely difficult to treat with conventional antibiotics, leading to a sharp rise in clinical complications.[1] Worldwide, they have been identified as a major nosocomial pathogen.[2,3] Along with the increase in the prevalence of resistant strains due to the abuse of antibacterial agents, there is an urgent need for a new class of antibacterial agents with an entirely different structure and mode of action.[4] During the last decade, various resistant mechanisms have increased worldwide in bacterial pathogens which lead to failure treatment in human and animal diseases.[5] These bacteria are able to adapt rapidly to new environmental conditions such as in the presence of antimicrobial molecules and as a consequence, resistance increases with the antimicrobial use.[6]

In recent years, various nanoparticles have been reported to display good biological activities.[7] Amongst them, TiO₂ nanoparticles have attracted the attention of researchers due to its oxidative and hydrolysis properties. As a photocatalyst, it can improve the efficiency of electrolytically splitting water into hydrogen and oxygen which can produce electricity in nanoparticles form. When these nanoparticles are exposed to ultraviolet (UV) light, they become increasingly hydrophilic.[8] In the case

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of antibacterial coatings, maximum density of nonmaterial on surface results in the generation of reactive oxygen species (ROS) and thus increases the antibacterial activity of TiO_2 . [9] TiO_2 photocatalysts have also been utilised to disinfect a broad spectrum of microorganisms. [10] Recently the metal oxide nanoparticles played a vital role in the novel drug delivery systems. The biosynthesised and chemically synthesised silver nanoparticles showed various biological activities. [11] Recent studies have demonstrated that specially formulated metal oxide nanoparticles have good antibacterial activity [12] and antimicrobial formulations comprising nanoparticles could be effective bactericidal materials. [13,14]

In the present study, the efficacy of TiO_2 nanoparticles as an antibacterial agent against MDR *P. aeruginosa* was explored. Also, the synergistic effect of these nanoparticles with the third generation cephalosporins, viz. cefotaxime (CEF) and ceftazidime (CEZ), was studied to see the enhancement in potency of these drugs.

2. Experimental

2.1. Strains and media

Four clinical isolates of *P. aeruginosa* were obtained from Sahyadri Hospital and Research Lab Pvt Ltd, Karve Road, Pune, Maharashtra, viz: Isolate 1 – Pus, Isolate 2 – Sputum, Isolate 3 – endo-tracheal tract (ETT), Isolate 4 – broncho-alveolar lavage (BAL). Control strain (NCIM-5029) of *P. aeruginosa* was obtained from National Chemical Laboratory, Pune. All solutions and reagents were prepared with distilled and deionised water and analytical grade chemicals were used without further purification (Sigma-Aldrich Co., MO, USA). Nutrient agar (NA) was used for the maintenance of clinical isolates of the bacterial strains.

2.2. Characterisation of nanoparticle

The optical property of TiO_2 nanoparticles was studied by ultraviolet-visible (UV-vis) spectroscopy. The crystal phase and crystallite size of TiO_2 nanoparticles were recorded by X-ray diffraction technique. In this $\text{CuK}\alpha$ radiation ($\lambda = 1.54 \text{ \AA}$) was used in the 2θ scan range of 10° – 80° . The full width at half maximum at the plane (101) was used to estimate the average crystallite size of TiO_2 .

2.3. Preparation of photocatalytic product

For testing antimicrobial activity in the standard TiO_2 nanoparticle film, 1 gram of commercial Degussa P25 TiO_2 powder was kept in vigorously shaking conditions to form aqueous solution. Then prepared solution was exposed to UV light for 0, 30 and 60 minutes.

2.4. Antibiotic susceptibility test

Antibiotic susceptibility test was carried out by ‘Disk Diffusion method’. [15,16] Mueller Hinton (MH) agar media was poured in 90 mm Petri dishes and incubated at

37 °C overnight to check sterility. The bacterial suspension was adjusted to an optical density (OD₅₄₅) according to McFarlands standard (10⁶ CFU/mL). Bacterial suspension was inoculated on each plate by spread plate method and then antibiotic sensitivity disks were placed on the surface of the medium. The plates were then incubated at 37 °C for 18 hours and sensitivity of each antibiotic was determined by checking the zones of inhibition.

2.5. Broth dilution assay

Sterile nutrient broth was poured in dilution tubes. Bacterial suspension was then added along with varying concentrations of TiO₂ nanoparticles (100–500 µg/mL) from the two stocks: (1) exposed to UV light for one hour and (2) no UV exposure. These samples are incubated for 24 hours at 37 °C and then minimum inhibitory concentration (MIC) value of TiO₂ nanoparticles was established. Here nanoparticles were used as a negative control while the respective strains of *P. aeruginosa* culture were taken as positive control.

2.6. Antibacterial assay

The antibacterial activity of the TiO₂ nanoparticles was checked by using well-diffusion method. Sterile MH agar was poured into petriplates and was swabbed with the overnight culture (10⁶ cells/mL) of bacteria. Then solid medium was gently punctured with the help of cork borer to make a well. After that, TiO₂ nanoparticle samples (25–1000 µg/mL) were added from the above-mentioned two stocks into each well and incubated for 24 hours at 37 °C and the antibacterial sensitivity was measured as the zone of inhibition in mm diameter. The same technique was used to establish the MIC values for the antibiotics CEF, CEZ and their combinations with TiO₂ nanoparticles.

2.7. Effect of UV irradiation on antimicrobial potential of TiO₂ nanoparticles

To check the effect of UV light exposure on antimicrobial potential of TiO₂ nanoparticles, three separate stocks were prepared using 50 µg TiO₂ nanoparticles were dissolved in distilled water and exposed to UV light (UV-C light – wavelength 254 nm) for 0, 30 and 60 minutes and used for antibacterial assay.[10,17]

3. Results and discussion

In the present study, the antibiotic sensitivity disk consisted of a total of 12 antibiotics viz: Amikasin, Ofloxacin, Gentamycin, Norfloxacin, Cefaclor, Ciprofloxacin, Nitrofuantoin, Cefoperazone, Nalidixic acid, Cefuroxime, Cefadroxil and Netlinocin were used against the selected four different isolates (Isolate 1 from Pus, Isolate 2 from Sputum, Isolate 3 from ETT and Isolate 4 from BAL) along with the control strain of *P. aeruginosa*. The antimicrobial susceptibility testing showed resistance in 5 out of 12 antibiotics for control and similarly 9, 8, 8 and 8 antibiotics for isolates 1, 2, 3 and 4, respectively, as shown in Table 1. This confirms *P. aeruginosa* as a MDR pathogen.

The commercially available standard Degussa-P25 TiO₂ nanoparticles have an average particle size of ~25 nm. It shows mixed rutile/anatase phase. The optical properties of

Table 1. Results of antibiotic susceptibility test against *P. aeruginosa*.

Antibiotics	Control strain	Isolate 1	Isolate 2	Isolate 3	Isolate 4
Amikasin AN	20 mm S	12 mm R	14 mm R	12 mm R	15 mm I
Ofloxacin OX	35 mm S	30 mm S	30 mm S	34 mm S	30 mm S
Gentamycin G	5 mm R	5 mm R	3 mm R	2 mm R	10 mm R
Norfloxacin NR	38 mm S	30 mm S	32 mm S	35 mm S	35 mm S
Cefaclor CFC	35 mm S	2 mm R	0 mm R	0 mm R	2 mm R
Ciprofloxacin Cip	45 mm S	35 mm S	43 mm S	37 mm S	40 mm S
Nitrofuantoin NF	2 mm R	0 mm R	0 mm R	0 mm R	0 mm R
Cefoperazone CFP	30 mm S	15 mm R	17 mm I	17 mm I	25 mm S
Nalidixic acid NA	0 mm R	0 mm R	0 mm R	0 mm R	0 mm R
Cefuroxime CR	10 mm R	0 mm R	0 mm R	0 mm R	0 mm R
Cefadroxil CD	0 mm R	0 mm R	0 mm R	2 mm R	0 mm R
Netlinocin NET	15 mm S	0 mm R	0 mm R	0 mm R	10 mm R

Note: S, susceptible; R, resistant; I, intermediate.

TiO₂ nanoparticles are studied by UV-vis spectroscopy which is shown in Figure 1. The figure clearly shows the absorption peak at ~430 nm with wide distribution. Figure 2 shows XRD pattern of Degussa-P25 TiO₂ nanoparticle which exhibits sharp reflections at 25.32°, 37.83°, 47.81°, 54.55° and 62.71° in the 2θ range of 10°–80°, which can be indexed as (101), (004), (200), (105) and (204). The calculated crystallite size was found to be ~6.0 nm by Scherer formula at the plane (101).

The commercial Degussa-P25 TiO₂ nanoparticle is the most widely used photocatalyst. [18] The irradiation of UV light on TiO₂ nanoparticle generates an electron hole pair on the TiO₂ surface. The hole in the valence band can react with H₂O or hydroxide ions adsorbed on the surface to produce hydroxyl radicals (OH[•]), and the electron in the conduction band can reduce O₂ to produce superoxide ions (O₂^{•-}). Both holes and OH[•] are extremely reactive with contacting organic compounds. Detection of other ROS, such as hydrogen peroxide (H₂O₂) and singlet oxygen, has also been reported.[10] Based on this information, the MDR strains of *P. aeruginosa* were then treated with TiO₂ nanoparticles and then exposed to UV light for 0, 30 and 60 minutes. According to Kim et al. in 2008, the optimal initial concentration of microorganism, light intensity and light irradiation time for the photocatalytic reaction of TiO₂ nanoparticles were 10⁶ CFU/mL, 1.0 mW/cm², 3 hours, respectively. The inoculum and light intensity were kept as above whereas the light irradiation time was varied. The MIC value of the nanoparticle was determined using broth dilution assay and confirmed by well-diffusion assay. No visible zone of

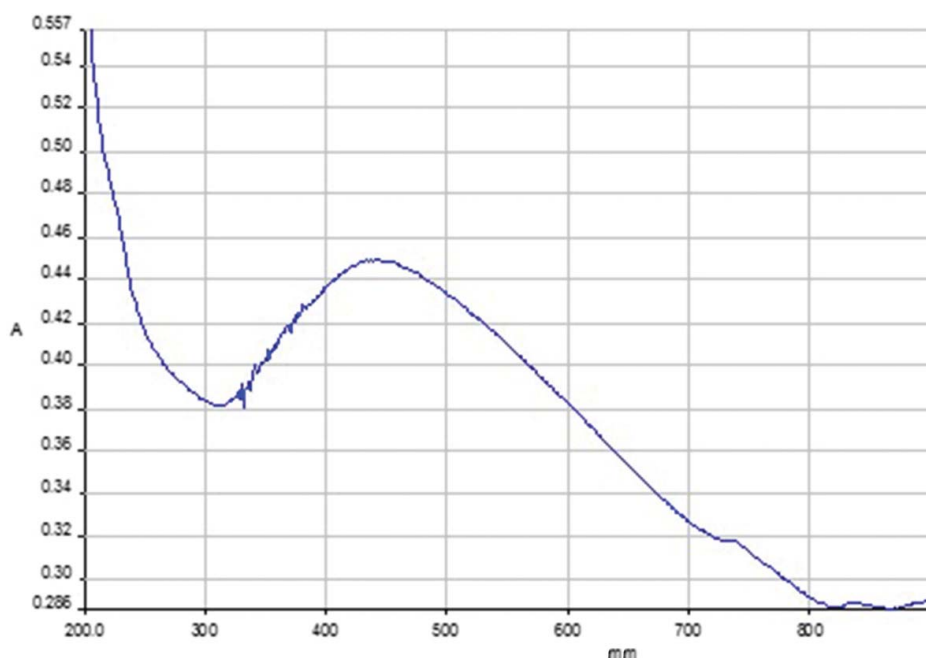


Figure 1. UV-vis absorption spectrum for TiO_2 nanoparticles.

inhibition was obtained for 0–500 $\mu\text{g/mL}$ concentrations of TiO_2 nanoparticle with 0 and 30 minutes UV irradiation. Further, with broth dilution assay, there was no change in turbidity for 0 and 30 minutes UV irradiation confirming the absence of antibacterial activity.

TiO_2 nanoparticle (0–500 $\mu\text{g/mL}$) when pre-exposed with one hour UV irradiation, showed antibacterial action against *P. aeruginosa*. MIC for TiO_2 nanoparticles was

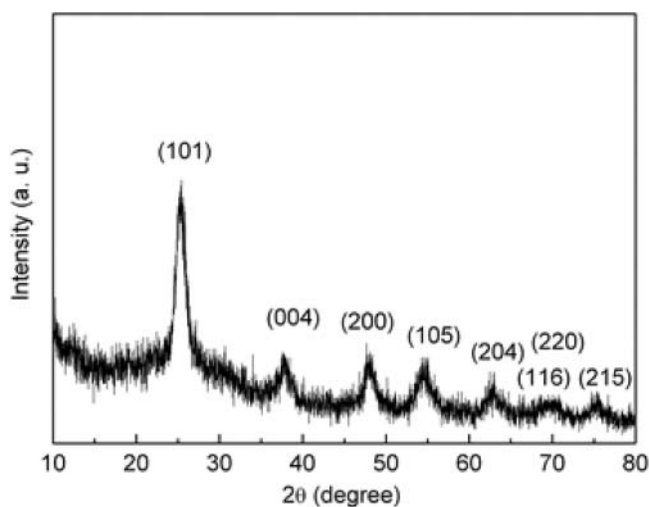


Figure 2. XRD pattern of crystalline TiO_2 nanoparticles.

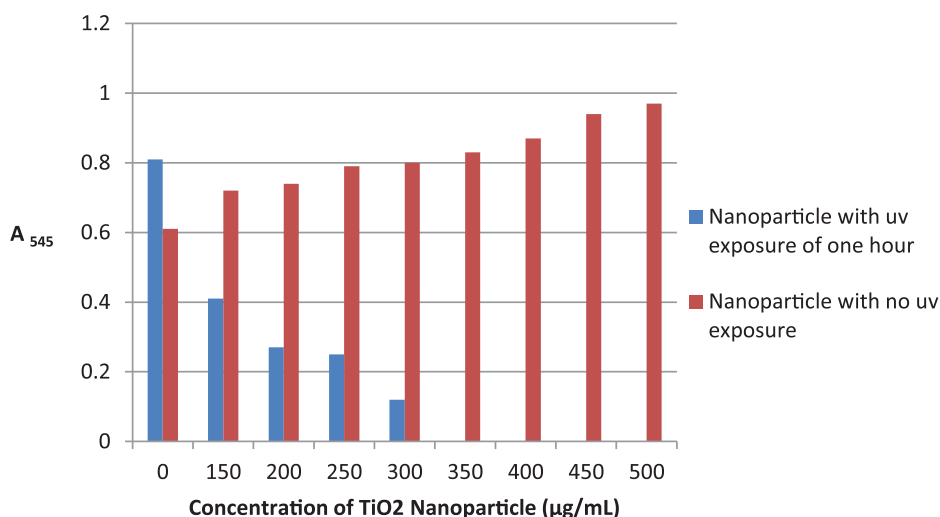


Figure 3. The effect of TiO₂ nanoparticles on the growth of *P. aeruginosa*.

obtained at 350 µg/mL as shown in Figure 3. An exposure to UV light on TiO₂ nanoparticles for 60 minutes may lead to the formation of ROS.[19] These ROS are known to bring about bacterial cell wall lysis, which is the fundamental reason for the antimicrobial activity of TiO₂ nanoparticles.[20] This antimicrobial activity is inferred on TiO₂ nanoparticles by UV light exposure for not less than one hour. At one-hour exposure, these nanoparticles obtained very strong inhibitory action against *P. aeruginosa* control strain, Isolate 1 (12 mm, zone of inhibition), Isolate 2 (14 mm, zone of inhibition), Isolate 4 (10 mm, zone of inhibition) and Isolate 3 (8 mm, zone of inhibition) at 350 µg/mL by well-diffusion assay.

MICs are most often used as a research tool to determine the *in vitro* activity of new antimicrobials and data from such studies have been used to determine MIC breakpoints. [21,22] In the present study MIC values for the antibiotics CEF and CEZ was found to be 130 µg/mL (13 mm, zone of inhibition) and 20 µg/mL (13 mm, zone of inhibition), respectively, against *P. aeruginosa*.

The antibacterial activity was not enhanced when TiO₂ nanoparticles was used in combination with the antibiotic CEF against *Pseudomonas* sp. On the other hand, the antibiotic CEZ as compared to antibiotic CEF showed an excellent synergistic effect (24 mm, zone of inhibition), when combined with TiO₂ nanoparticles as shown in Figure 4. The other four isolates also showed the similar inhibitory effect.

TiO₂ nanoparticles when used in combination with the antibiotic CEZ thus can act as a potential antibacterial agent which is shown in Figure 5. The change in MIC of CEZ in combination with TiO₂ nanoparticle can be attributed to the fact that either the sensitivity of antibiotic is enhanced by the presence of nanoparticles or none of the nanoparticle/CEZ in combination is suppressed.

The increasing frequency of MDRPA strains is a matter of concern as efficacious antimicrobial options are severely limited. As known nosocomial infections caused by *P.*

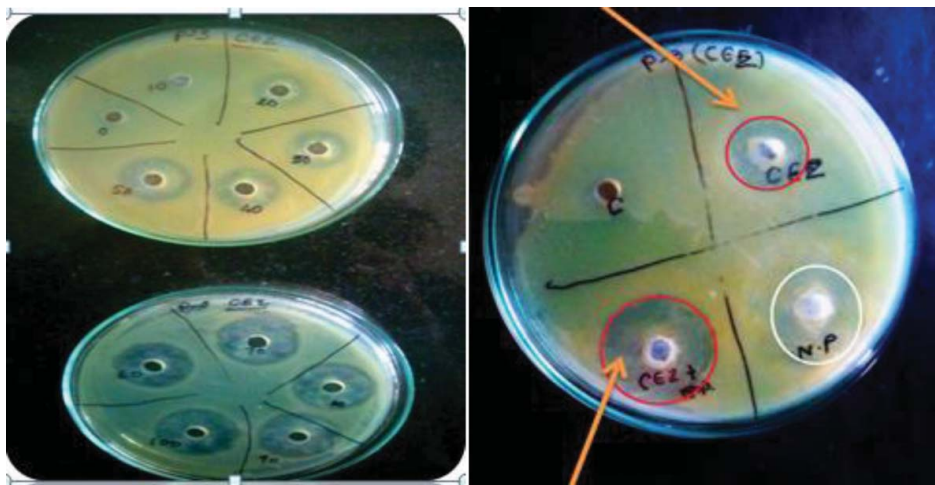


Figure 4. (Colour online) Determination of MIC of Ceftazidime (10–100 $\mu\text{g/mL}$) against multi-drug resistant *P. aeruginosa*. On the right, arrows showing CEZ alone (13 mm, zone of inhibition) and CEZ + nanoparticle (24 mm, zone of inhibition).

aeruginosa are frequently life threatening and often challenging to treat. One potential strategy is to target the regulation of bacterial resistance mechanisms as a pathway to enhance the potency of available drugs and perhaps, restore the efficacy of available drugs.[23] In such conditions, combination therapy might prove useful for countering drug resistance.

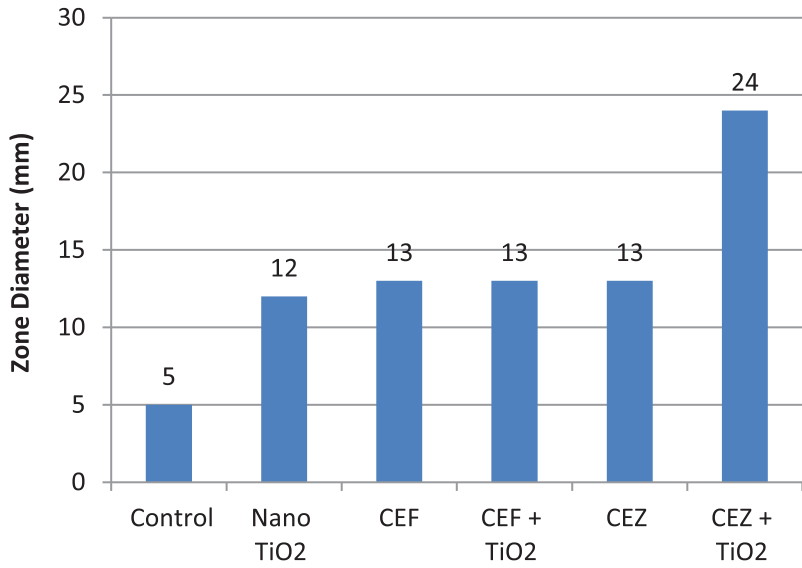


Figure 5. Effect of TiO_2 nanoparticles alone (350 $\mu\text{g/mL}$), ceftazidime (CEZ) alone (20 $\mu\text{g/mL}$), cefotaxime (CEF) alone (130 $\mu\text{g/mL}$), CEF and TiO_2 nanoparticle in combination, CEZ and nanoparticle on multi-drug resistant *P. aeruginosa* shown as zone diameter (mm). For control 5 mm diameter shown is the well diameter.

4. Conclusion

In the current scenario, multi-resistant infectious disease organisms are difficult and sometimes impossible to treat successfully. Further no new antimicrobial agents, active against MDR strains of *P. aeruginosa*, are in advanced stages of development as therapeutic options. In the present study, an exposure of UV irradiation of 60 minutes has shown to greatly enhance the antibacterial efficacy of TiO₂ nanoparticle against MDR *P. aeruginosa*. Further, the antibiotic CEF potency was increased when used in combination with 60 minutes UV-irradiated TiO₂ nanoparticles. This synergistic study against the pathogens offers its application potential in clinical diagnostics as an attempt to find a solution to the growing problem of antibiotic resistance. This approach is promising and thus can broaden the repertoire of useful anti-pseudomonal drugs.

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